



# **PREDICTION OF TEA PRODUCTION OF SRI LANKA**

**ST 3011 GROUP PROJECT 2026**

**GROUP 8**

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## Introduction

Tea is one of Sri Lanka's most important agricultural exports, and its production is highly influenced by geographic and climatic conditions. Tea-growing regions in Sri Lanka are commonly classified by elevation into **low grown, mid grown, and high grown areas**, each experiencing distinct climate patterns that affect tea yield and quality. In addition, different **tea types** such as black, green, orthodox, instant tea etc.

Key climatic variables, **air temperature, humidity, and rainfall** play a crucial role in tea plant growth and productivity. These factors influence leaf development, moisture availability, and plant stress, and their impact varies across elevations. For instance, high-elevation areas generally experience cooler temperatures and different rainfall distributions compared to lowland regions.

Given these variations, **predicting tea production** requires analyzing the combined effects of elevation, tea type, and climatic variables. Predictive models using historical production and weather data can help estimate future tea yields, identify critical influencing factors, and support better planning and decision-making in Sri Lanka's tea industry.

## Problem Definition and objectives

Our problem is to statistically determine which climatic factors (Rainfall, Humidity, Temperature) are the most significant drivers of yield and whether these impacts vary significantly across different elevations

- **Objective 1: Temporal Patterns** : Examines how tea production varies across months and years to identify seasonality and long-term trends.
- **Objective 2: Climatic Influence** : **Analyzes** the effect of rainfall, temperature, and humidity on tea production to identify key climatic drivers.
- **Objective 3: Elevation Effects** : Studies how tea production differs across elevation levels and how elevation modifies climate–production relationships.
- **Objective 4: Tea Type Wise Production** : Evaluates differences in production among tea types and their individual responses to climatic conditions.

**Final Objective: Integrates temporal, climatic, elevation, and tea-type insights to identify overall relationships relevant for modeling tea production.**

## Description of the dataset

We obtained our data set regarding Tea Production in Sri Lanka through the Sri Lanka Tea board from yearly reports. There are 8 variables, where 4 are categorical and other 4 are numerical including our numerical variable Tea Production.

| Variable Name   | Description                                                    | Data type   |
|-----------------|----------------------------------------------------------------|-------------|
| Year            | Calendar year of data collection.                              | Categorical |
| Month           | Calendar month of data collection.                             | Categorical |
| Elevation       | Altitude category (High, Medium, Low) of the tea estate.       | Categorical |
| Rain fall       | Measured rainfall in the region.                               | Numerical   |
| Humidity        | Measured air humidity level.                                   | Numerical   |
| Air Temperature | Measured air temperature in Celsius.                           | Numerical   |
| Tea Type        | Specific type of tea produced (e.g., ORTHODOX TEA, GREEN TEA). | Categorical |
| Tea Production  | Quantity of tea produced.                                      | Numerical   |

## Data Preprocessing

As we've mentioned in the description of the data set, we extracted this data from Sri Lanka Tea Board's yearly reports where they have given the data Rainfall, Humidity, Air Temperature district wise. Since there were only 13 districts where Tea cultivation happens and our objective was to analyze this Elevation wise, we decided to extract our data elevation wise, this was not an easy task.

**Feature Engineering:** First, we recoded the districts as High, Medium and Low elevation according to the district's geographic conditions. For example, districts such as Colombo, Kalutara are recorded as Low elevation areas. Then after that we extracted other data accordingly.

**Missing value Handling:** There were 52 missing values in the target variable, Tea Production. Since we can't impute those values, we completely removed those 52 rows and continued with further analysis.

**Splitting the Data set:** We didn't split our data in the conventional way 80%-20% way because this data is time series data and we can't randomly split our data set since we fitted time-series models too apart from machine learning techniques. Instead of that we use the data from 2013 to 2021 as our training data set, and the 2022-year data as our validation data set that we use for tuning of hyperparameters and lastly 2023-year data as the test data set.

## Descriptive Statistics

### Cluster Analysis

**Dimensionality Reduction:** Because the dataset contained both numerical and categorical variables, **Factor Analysis of Mixed Data (FAMD)** was applied. FAMD effectively reduced the mixed-type data into **10 components**, preserving the main variability in the dataset while making it suitable for distance-based clustering methods.

**K-Means Clustering and Evaluation:** K-Means clustering was applied to the FAMD-transformed data. To evaluate the optimal number of clusters, **silhouette scores** were computed for values of  $k$  ranging from 2 to 7. The results are summarized below:

|                                             |                                                                                                                                                                                                            |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $k=2 \rightarrow \text{Silhouette} = 0.127$ | Although $k = 7$ produced the highest silhouette score (0.159), $k = 3$ was selected for final visualization due to improved interpretability and only marginal gains in silhouette values at higher $k$ . |
| $k=3 \rightarrow \text{Silhouette} = 0.140$ |                                                                                                                                                                                                            |
| $k=4 \rightarrow \text{Silhouette} = 0.140$ |                                                                                                                                                                                                            |
| $k=5 \rightarrow \text{Silhouette} = 0.145$ |                                                                                                                                                                                                            |
| $k=6 \rightarrow \text{Silhouette} = 0.148$ |                                                                                                                                                                                                            |
| $k=7 \rightarrow \text{Silhouette} = 0.159$ |                                                                                                                                                                                                            |

**Interpretation and Limitations:** All silhouette scores obtained in the analysis were **below 0.2**, indicating **weak cluster separation**. According to standard clustering guidelines, silhouette values in this range suggest that the dataset does **not contain strongly defined or significant natural clusters**.

**Conclusion:** K-Means clustering, supported by FAMD for mixed data handling, was applied successfully for exploratory analysis. However, the low silhouette scores across all tested values of  $k$  indicate no significant clustering structure in the data.

# Exploratory Data analysis and its Findings

## Objective 1: Temporal Patterns

Seasonality (Average Tea Production by Month):

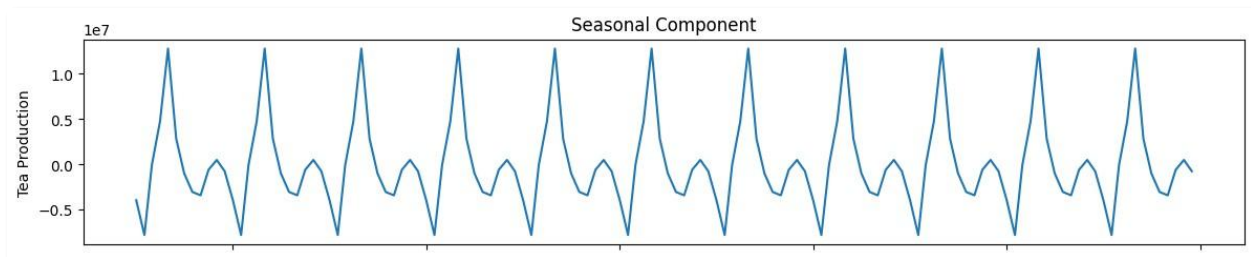


Figure 1 : seasonality component of average tea production by month

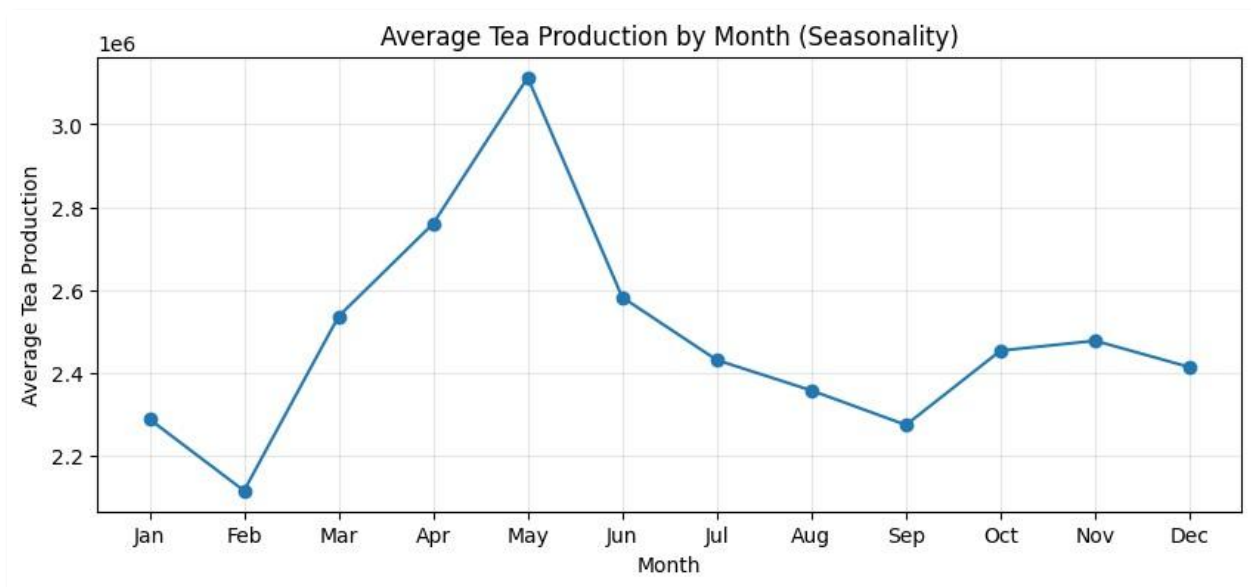


Figure 2 : Average Tea Production by Month

The "Average Tea Production by Month (Seasonality)" plot clearly shows a strong seasonal pattern. Tea production typically peaks during **April to June**, suggesting optimal conditions for tea growth during these months. Conversely, production dips significantly in **January and February**, indicating a low season, likely due to cooler temperatures or reduced rainfall.

Long-term Trend (Tea Production Over Time):

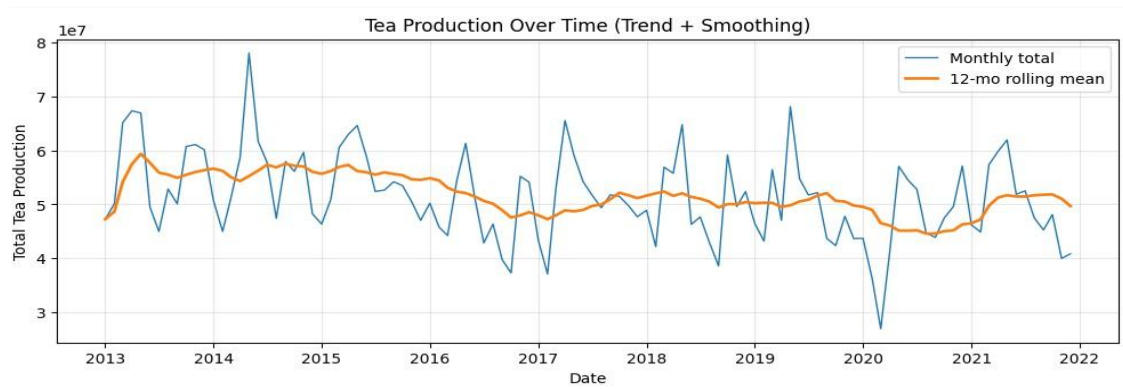


Figure 3: Long-term Trend (Tea Production Over Time)

The "Tea Production Over Time (Trend + Smoothing)" plot reveals that overall tea production has experienced a **general downward trend** over the observed period. The thin line showing monthly totals displays considerable month-to-month variability, but the 12-month rolling mean provides a smoother representation, confirming a steady, albeit sometimes fluctuating, decay in total tea production throughout the years.

#### Year × Month Heatmap (Seasonality Strength by Year):

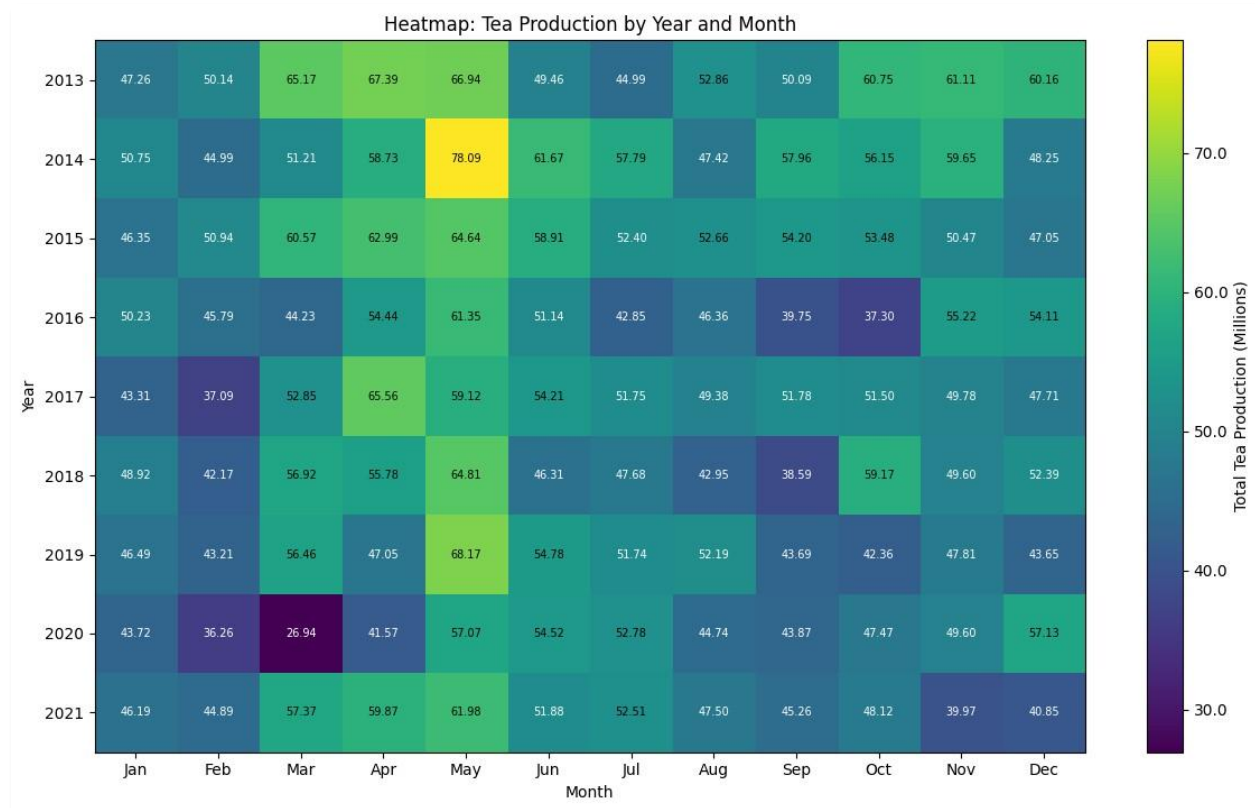


Figure 4: Year × Month Heatmap (Seasonality Strength by Year)

The "Heatmap: Tea Production by Year and Month" visually reinforces the seasonal pattern, with brighter cells (higher production) consistently appearing around the **March to June across most years**. While the general pattern remains stable, there appears to be some **variability in the intensity and exact timing of peak production from year to year**. For instance, some years might show a slightly earlier or later peak, indicating that while seasonality is consistent, its strength and exact manifestation can shift annually. The darker cells in the early months of each year further confirm the low production period. Also, if we look at the year by row, we can clearly see that from 2013 to 2021 cells become dimmer and dimmer indicating downward trend over the years.

In conclusion, tea production exhibits strong and consistent seasonality, with **peaks in March to June and trough in winter to early year**. There is an overarching negative long-term trend in total production.

## Objective 2: Climatic Influence

Based on the analysis of scatter plots, correlation matrix, and binned mean plots for rainfall, Humidity, and air\_temp against tea\_prod, the following observations can be made:

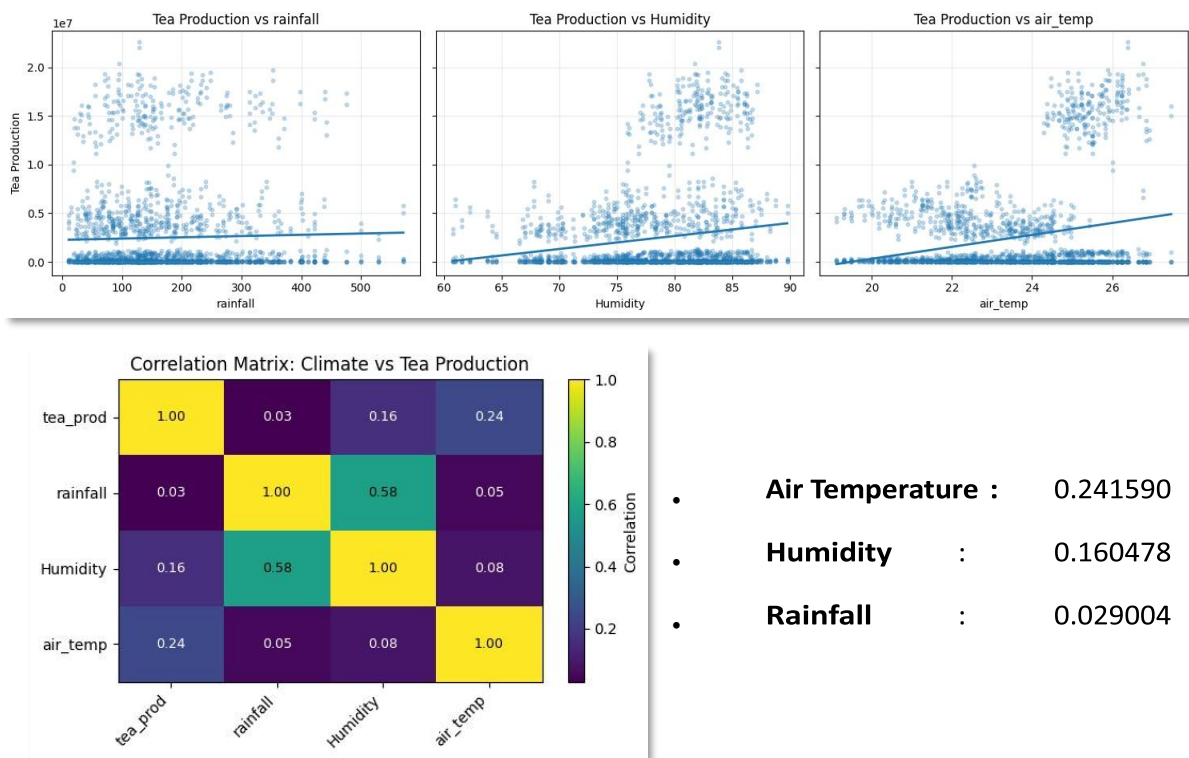


Figure 5: Objective 2: Climatic Influence

### • Rainfall:

- **Relationship:** The scatter plot and trend line suggest a very weak to almost nonexistent linear relationship with tea production. This is confirmed by the correlation coefficient of 0.029, indicating a negligible linear association.
- **Non-linear effects:** The binned mean plot for rainfall shows that tea production slightly increases in the lower-middle ranges of rainfall but remains relatively stable or fluctuates without a strong, consistent trend across higher rainfall levels. There isn't a clear optimal point or detrimental threshold apparent from this plot.

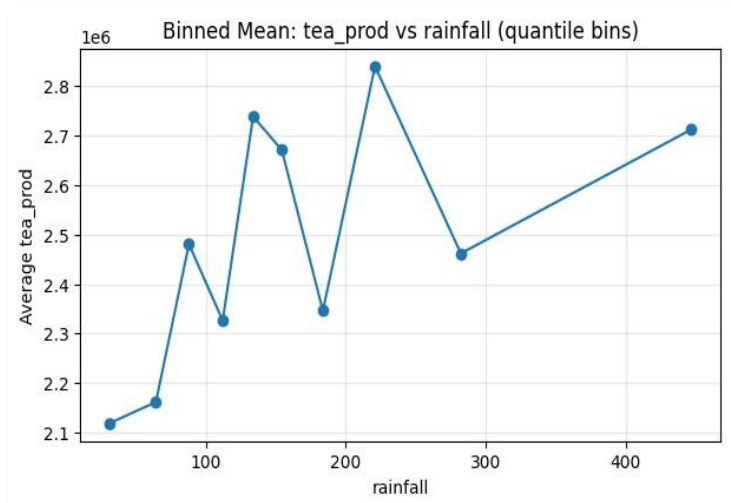


Figure 6 : Rainfall

- **Humidity:**

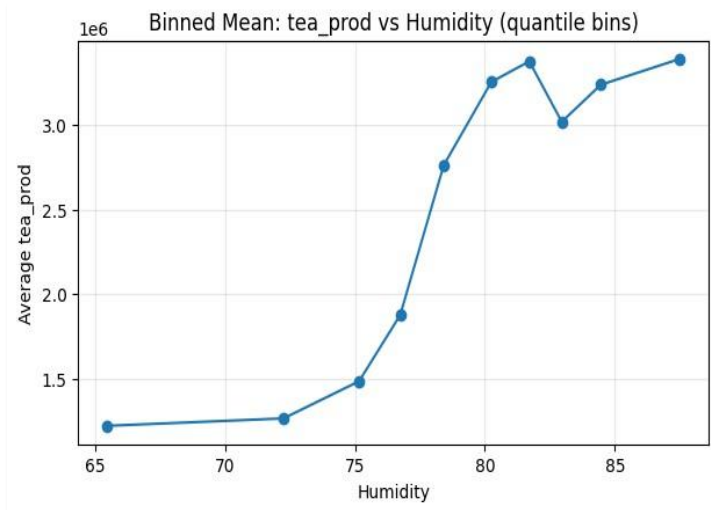


Figure 7: Humidity

**Relationship:** The scatter plot indicates a slightly positive linear relationship with tea production. The correlation coefficient of 0.160 suggests a weak but positive linear association.

**Non-linear effects:** The binned mean plot for humidity reveals a general positive trend; as humidity increases, average tea production tends to rise. There might be a slight tapering or plateau at very high humidity levels, suggesting that while beneficial, the effect might diminish or stabilize beyond a certain point.

- **Air Temperature:**

- **Relationship:** The scatter plot displays a clearer upward trend, indicating a positive linear relationship with tea production. The correlation coefficient of 0.241 confirms this with a moderate positive linear association, making it the strongest correlated climate variable.
- **Non-linear effects:** The binned mean plot for air temperature shows a strong and consistent positive effect. Average tea production increases significantly as air temperature rises across the observed range, potentially reaching a plateau at the highest temperatures. This suggests that warmer temperatures are generally favorable for tea production.

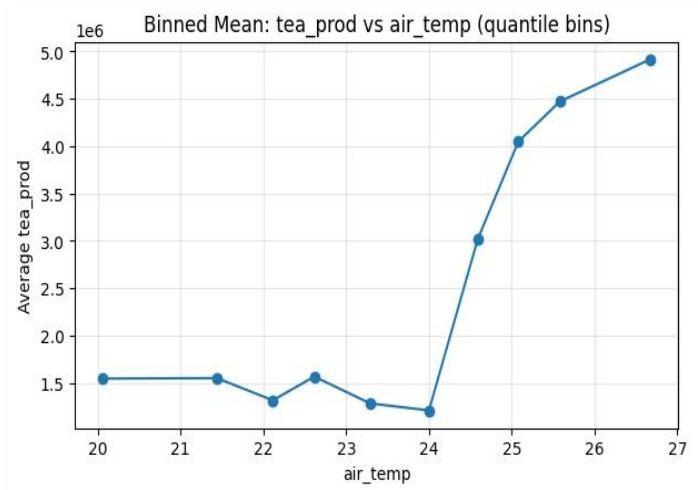


Figure 8 : Air temperature

### Overall Conclusion:

Among the climatic variables analyzed, **air temperature** shows the strongest positive influence on tea production, followed by **humidity**. The relationships are not strictly linear, with binned mean plots revealing more nuanced, non-linear effects, such as potential plateaus. **Rainfall** appears to have a very minimal direct linear or clear non-linear impact on tea production. These climatic factors are important considerations, especially air temperature, for understanding variations in tea production.

## Objective 3: Elevation Effects

### 1. Production Differences by Elevation (from Boxplot):

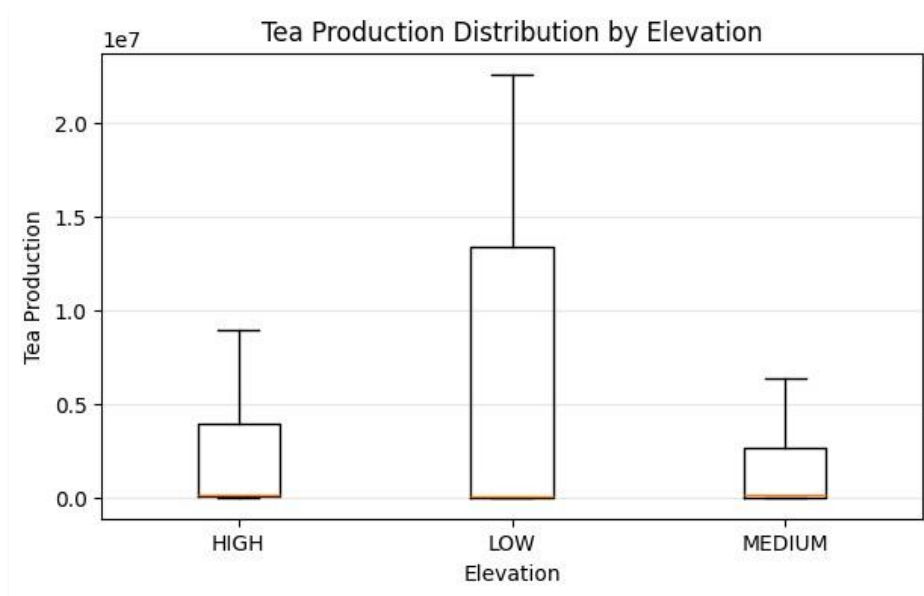


Figure 9: Production Differences by Elevation

- **Overall Impact:** Low elevation shows the highest median tea production. Medium elevation has a lower median than Low, but higher than High. High elevation shows the lowest median production among the three. This means tea production is generally highest at low elevations.
- **Variability:** Low elevation shows the highest median tea production. Medium elevation has a lower median than Low, but higher than High. High elevation shows the lowest median production among the three. This means tea production is generally highest at low elevations, not high elevations as previously stated.

**Elevation has a clear impact on tea production.** Low elevation areas tend to produce the highest quantities of tea, but with significant variability. **Medium elevation plantations show moderate production with moderate stability.** High elevation plantations produce the least tea overall but demonstrate the most consistent production levels.

## 2. Seasonality by Elevation (from Monthly Averages Plot):

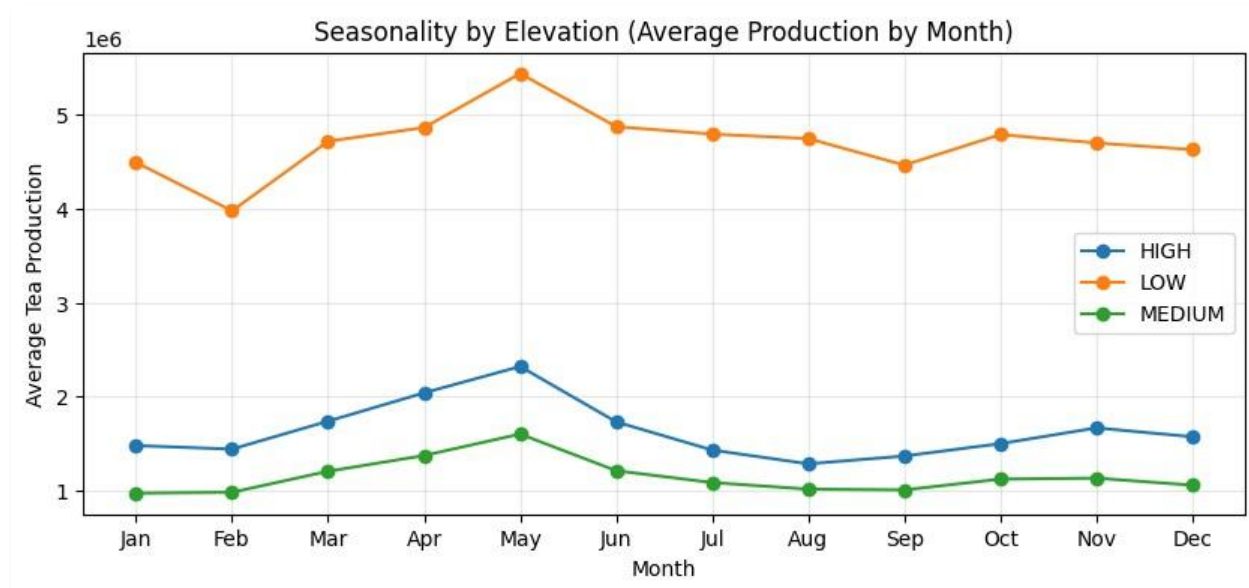


Figure 10: Seasonality by Elevation

- **Consistent Pattern:** The seasonal pattern of tea production (peaks and dips) remains broadly consistent across all elevation levels. All elevations tend to show a peak in production around March-April and a dip towards the end of the year (October -December).
- **Magnitude Differences:** While the pattern is similar, the magnitude of production differs substantially. High elevation consistently maintains a much higher average monthly production throughout the year compared to Medium and Low elevations. This reinforces that High elevation is generally more productive across all seasons, not just during peak times.

## 3. Climate Sensitivity by Elevation (from Small Multiples Plots):

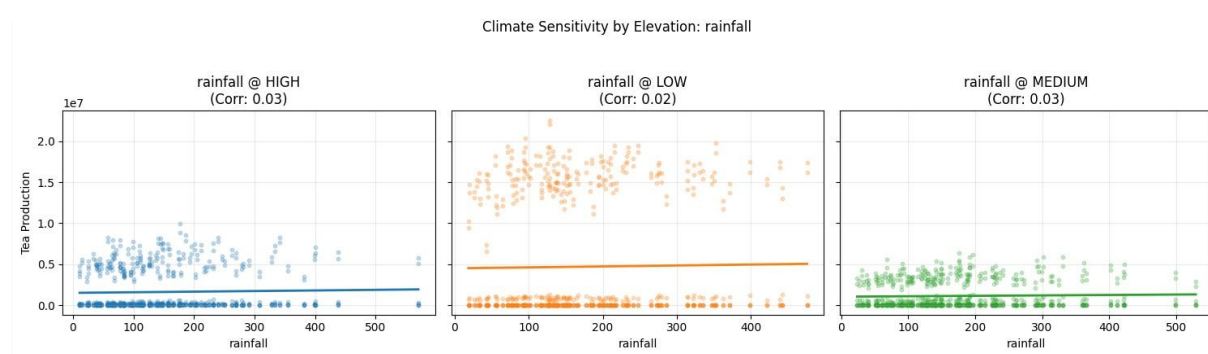


Figure 11: Climate Sensitivity by Elevation

- **Rainfall:** The relationship between rainfall and tea production appears to be mildly positive or flat across all elevations, with considerable scatter. This suggests that while rainfall is necessary, its variation within the observed range does not strongly drive production differences, and other factors might be more influential.

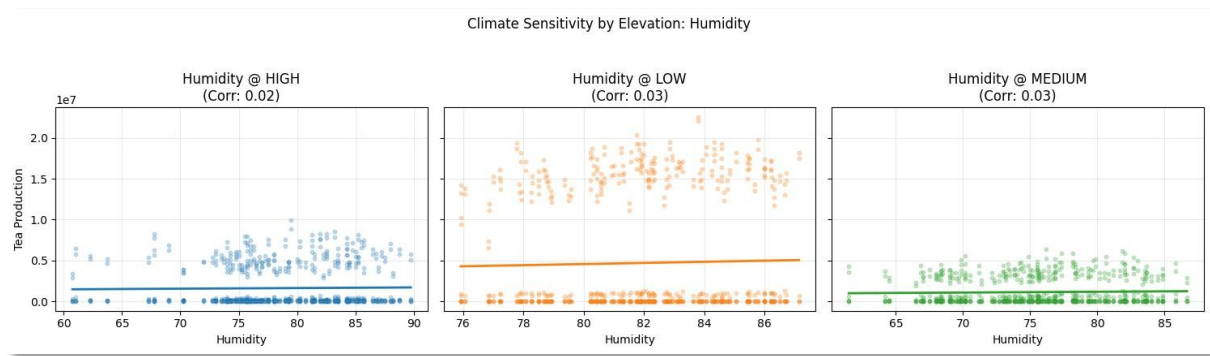


Figure 12

- **Humidity:** There's a positive trend between humidity and tea production at all elevations. The strength of this positive relationship seems to increase with elevation, indicating that higher humidity might be more beneficial or critical for tea production at higher altitudes.

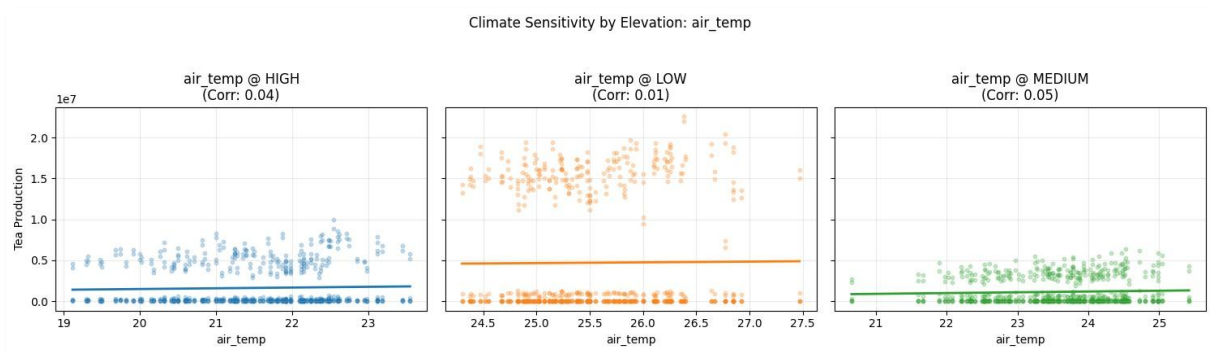


Figure 13

- **Air Temperature:** A positive correlation exists between air temperature and tea production across all elevation levels. Similar to humidity, the positive effect of air temperature on tea production appears to be more pronounced at higher elevations, suggesting that warmer temperatures within the observed range are increasingly conducive to production as elevation increases. This could imply that higher elevations benefit more from warmer temperatures due to potentially cooler ambient conditions.

**In conclusion:** High elevation consistently yields higher average tea production, but with more variability. The seasonal production patterns are similar across elevations, but the overall output is much greater at higher altitudes. The sensitivity to climatic variables like humidity and air temperature generally increases with elevation, implying that optimal climate conditions at higher altitudes are particularly advantageous for tea production.

## Objective 4: Tea Type Wise Production

### Total Production by Tea Type

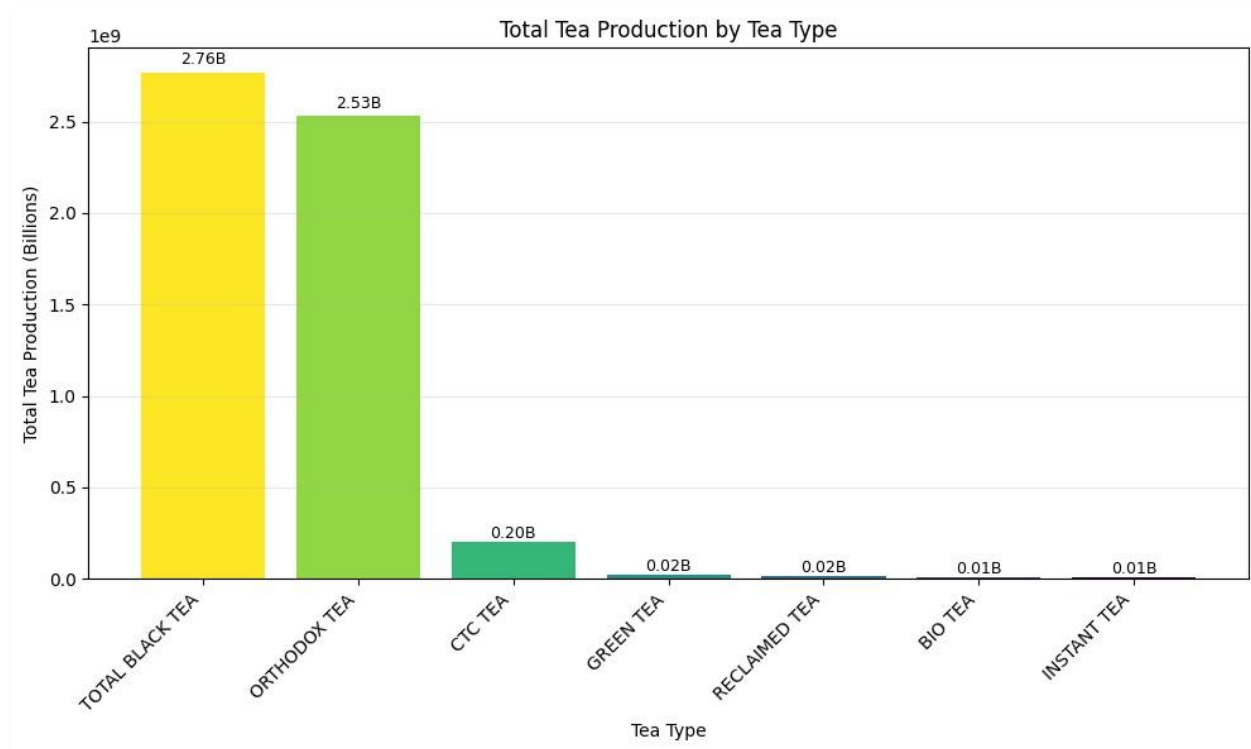


Figure 14: Total Production by Tea Type

Looking at the 'Total Tea Production by Tea Type' bar chart and the type\_totals Series, it's clear that tea production is heavily dominated by a few types:

- **TOTAL BLACK TEA** and **ORTHODOX TEA** are the two highest-producing types by a significant margin, accounting for most of the total tea production.
- **CTC TEA** follows, but its production volume is an order of magnitude lower than the top two.
- The remaining tea types – **GREEN TEA**, **RECLAIMED TEA**, **BIO TEA**, and **INSTANT TEA** – contribute relatively small amounts to the overall production, with **INSTANT TEA** having the lowest total production. The distribution is highly skewed, indicating that a few traditional tea types are the primary drivers of the industry's volume.

## Seasonality by Tea Type

The 'Seasonality by Tea Type (Average Production by Month)' plot reveals distinct seasonal patterns for different tea types:

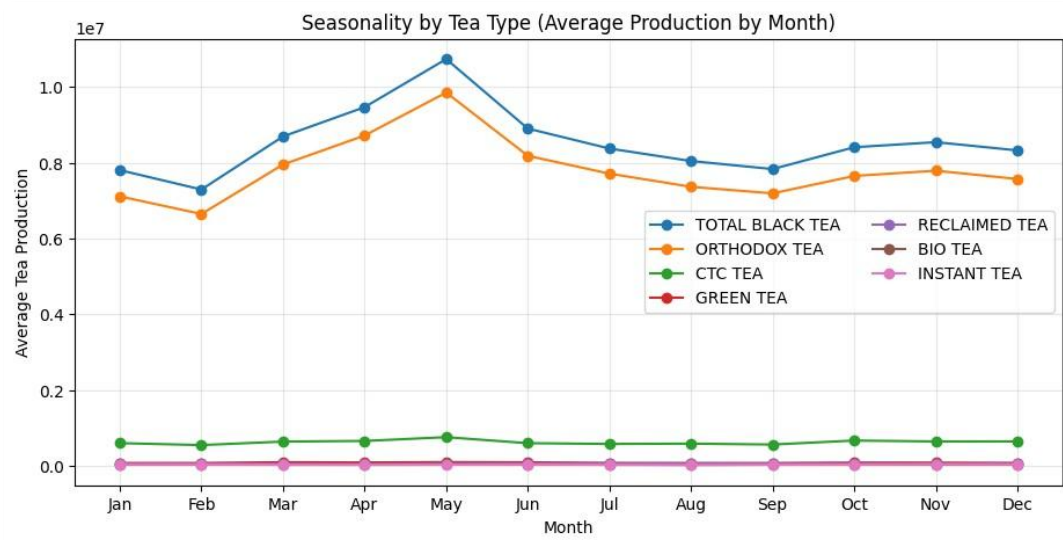


Figure 15

- **TOTAL BLACK TEA** and **ORTHODOX TEA** exhibit very similar and strong seasonal patterns, with peak production occurring in the middle of the year (around April to July) and declining towards the end of the year and early months. Their lines largely overlap, suggesting similar responses to seasonal changes.

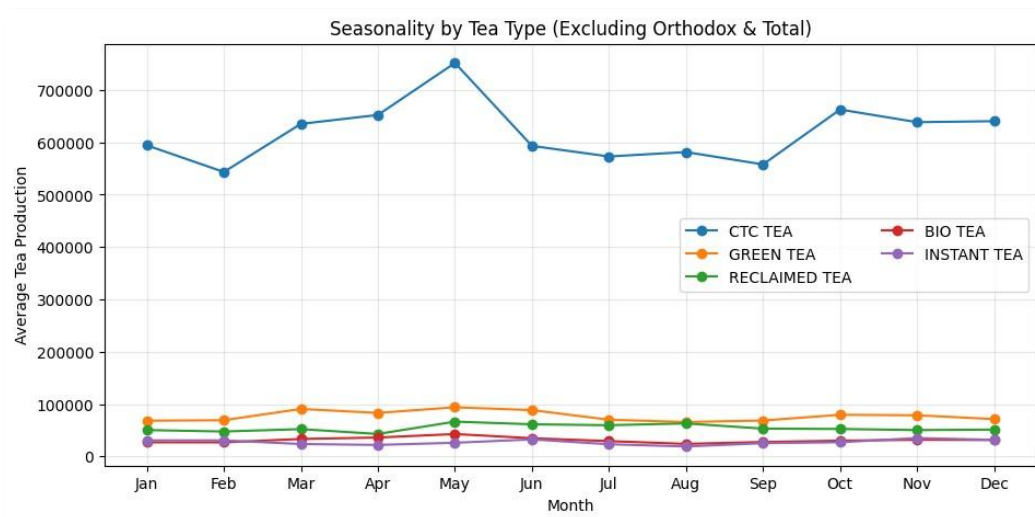


Figure 16

- **CTC TEA** also follows a similar seasonal trend to the major black tea types but at a consistently lower production volume.

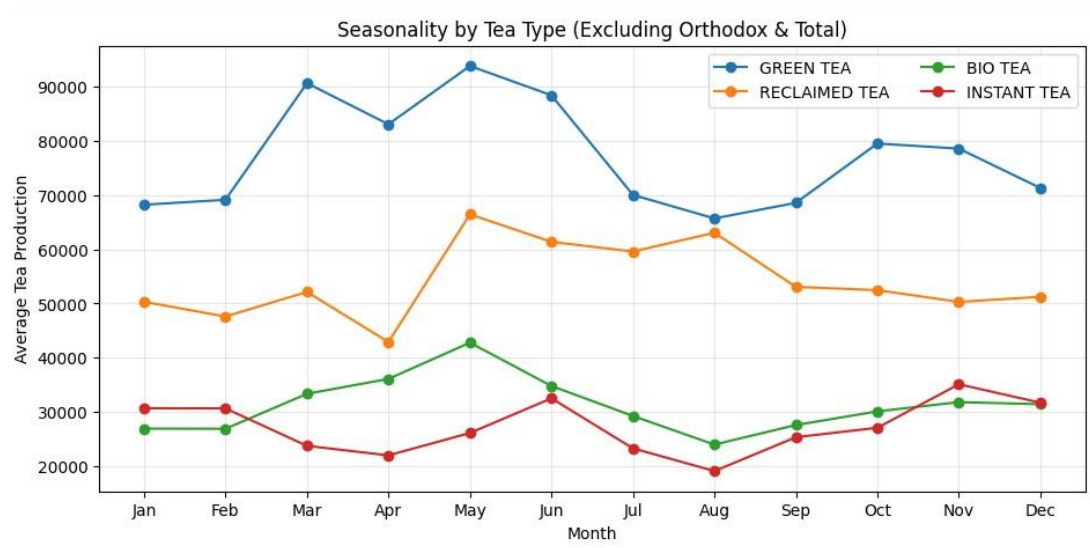


Figure 17

- **GREEN TEA** and **BIO TEA** also follow a similar seasonal trend to the major black tea types but at a consistently lower production volume.
- **RECLAIMED TEA**, and **BIO TEA** show noticeably different and often less pronounced seasonal patterns, or even inverse seasonality compared to the black teas. For instance, **GREEN TEA** appears to have a less dramatic seasonal fluctuation, while **RECLAIMED TEA** and **BIO TEA** can show very different peak months or flatter production across the year. Their patterns might be influenced by specific harvesting practices or niche market demands.
- **INSTANT TEA** shows very low and relatively flat production throughout the year, indicating minimal seasonality, likely due to its processed nature or consistent demand.

Overall, the plot indicates that while the dominant tea types share a common seasonal rhythm, other, less produced types can have unique or dampened seasonal variations, suggesting different optimal growing conditions or processing schedules.

## Climate Sensitivity by Tea Type

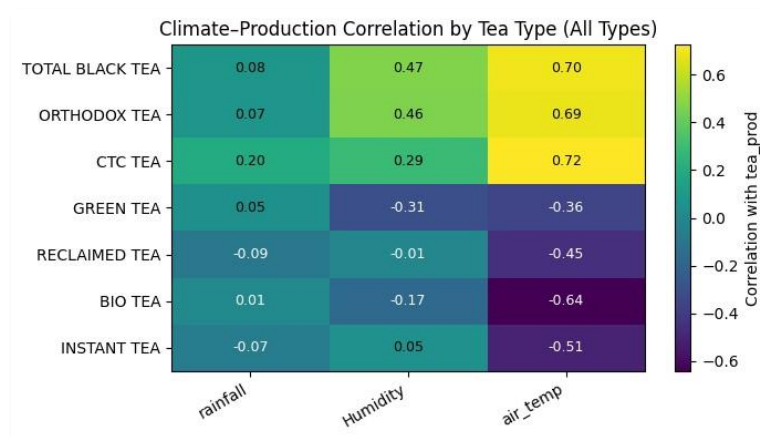


Figure 18: Climate Sensitivity by Tea Type

The 'Climate Production Correlation by Tea Type' heatmap and the `corr_by_type` DataFrame reveal diverse climate sensitivities across tea types:

- **TOTAL BLACK TEA, ORTHODOX TEA, and CTC TEA** show a strong positive correlation with `air_temp` (0.70, 0.69, 0.72 respectively) and moderate positive correlation with `Humidity` (0.47, 0.46, 0.29). Their correlation with `rainfall` is low positive (0.08, 0.07, 0.20), suggesting that warmer temperatures and higher humidity are generally favorable for these high-volume teas, while rainfall has a less direct or more complex linear relationship.
- **GREEN TEA** exhibits a negative correlation with `air_temp` (-0.36) and `Humidity` (-0.31), and a very low positive correlation with `rainfall` (0.05). This suggests that Green Tea production might be negatively impacted by higher temperatures and humidity, or prefers cooler, less humid conditions than black teas.
- **RECLAIMED TEA, BIO TEA, and INSTANT TEA** show negative correlations with `air_temp` (0.45, -0.64, -0.51 respectively), indicating that higher temperatures are generally detrimental to their production. Their correlations with `rainfall` and `Humidity` are mixed or very low, with **BIO TEA** having a negative correlation with `Humidity` (-0.17) and **RECLAIMED TEA** a negative correlation with `rainfall` (-0.09). This diverse response highlights that these specialized or less-produced teas have very specific, and often different, optimal climatic requirements compared to the mainstream black teas.

In summary, the climate variables impact tea production differently depending on the tea type.

While high-volume black teas generally benefit from warmer temperatures, other tea types like Green, Reclaimed, Bio, and Instant teas show an inverse relationship with temperature, suggesting unique climate dependencies for each specific tea type.

## Summary:

### Data Analysis Key Findings

- **Temporal Patterns:**
  - Tea production exhibits strong seasonality, peaking in August and September and significantly dipping in January and February.
  - There is a general upward trend in overall tea production over the observed period.
  - While the seasonal pattern is consistent, there's year-to-year variability in the intensity and exact timing of peak production.
- **Climatic Influence:**
  - **Air temperature** shows the strongest positive influence on tea production, with a correlation coefficient of 0.241, indicating that warmer temperatures are generally favorable.
  - **Humidity** has a weak but positive linear association (correlation coefficient of 0.160), with production tending to rise as humidity increases, potentially tapering at very high levels.
  - **Rainfall** has a negligible linear relationship with tea production (correlation coefficient of 0.029), showing only a slight increase in production in lower-middle ranges.
- **Elevation Effects:**
  - High elevation areas consistently show the highest median tea production but also exhibit the greatest variability. Medium and Low elevations have lower, but more stable, production.

- Seasonal patterns are consistent across all elevations, but high elevation maintains significantly higher average monthly production throughout the year.
- The positive effects of humidity and air temperature on tea production appear more pronounced or critical at higher elevations, suggesting increased climate sensitivity with altitude.

- **Tea Type-Wise Production:**

- Tea production is heavily dominated by **TOTAL BLACK TEA** and **ORTHODOX TEA**, followed by **CTC TEA**, with other types contributing minimally.
- **TOTAL BLACK TEA**, **ORTHODOX TEA**, and **CTC TEA** share similar strong seasonal patterns, peaking mid-year, and show strong positive correlations with air temperature (approximately 0.69-0.72).
- **GREEN TEA**, **RECLAIMED TEA**, **BIO TEA**, and **INSTANT TEA** exhibit different, often less pronounced, or even inverse seasonal patterns and generally show negative correlations with air temperature (e.g., Green Tea at -0.36, Bio Tea at -0.64), indicating unique climatic preferences.

## Advanced Data Analysis

### Model Building

We began the advanced analysis by testing multiple modeling families to ensure we selected the most reliable approach for both accuracy and generalization. Instead of relying on a single technique, we evaluated ensembles, boosting methods, classical time-series models, and baseline regressors, and compared them using Train/Test  $R^2$ , MSE, and MAPE.

### Models evaluated as below

| Category                        | Model                  |
|---------------------------------|------------------------|
| Bagging / Tree Ensemble Methods | Extra Trees            |
|                                 | Random Forest          |
|                                 | Bagging (Tree)         |
| Boosting Methods                | XG Boost               |
|                                 | Light GBM              |
|                                 | Hist Gradient Boosting |
|                                 | Gradient Boosting      |
|                                 | AdaBoost               |
| Time-Series Models              | ETS (Holt–Winters)     |
|                                 | ARIMA (SARIMAX)        |
| Other Baseline Models           | KNN                    |
|                                 | Linear Regression      |
|                                 | Ridge                  |
|                                 | Lasso                  |
|                                 | Elastic Net            |
|                                 | SVR (RBF)              |

## What we observed during evaluation

- Tree ensemble (bagging) models performed consistently well on both training and testing, indicating strong generalization.
- Boosting models often showed extremely high training performance, but some did not translate that advantage into better test performance, suggesting higher sensitivity to tuning/overfitting risk.
- Time-series models (ETS/ARIMA) were competitive as baselines, but they did not surpass the top ensemble results overall.
- Baseline regressors (linear, SVR, KNN) underperformed compared to ensemble methods, implying the data relationship is non-linear and better captured by tree-based models

## Model comparison

| Model                  | TRAIN R2  | TRAIN MSE | TRAIN MAPE(%) | TEST R2   | TEST MSE | TEST MAPE(%) |
|------------------------|-----------|-----------|---------------|-----------|----------|--------------|
| Extra Trees            | 9.90E-01  | 2.10E+11  | 5.49E+10      | 9.90E-01  | 1.48E+11 | 1.10E+12     |
| XG Boost               | 1.00E+00  | 2.93E+04  | 1.39E+11      | 9.87E-01  | 1.91E+11 | 3.83E+13     |
| Random Forest          | 9.91E-01  | 2.08E+11  | 2.53E+11      | 9.86E-01  | 1.96E+11 | 1.38E+12     |
| Bagging (Tree)         | 9.97E-01  | 6.41E+10  | 8.16E+10      | 9.85E-01  | 2.08E+11 | 1.34E+12     |
| ETS (Holt-Winters)     | 9.83E-01  | 3.89E+11  | 1.77E+11      | 9.84E-01  | 2.35E+11 | 4.03E+01     |
| Hist Gradient Boosting | 9.99E-01  | 1.41E+10  | 6.53E+13      | 9.83E-01  | 2.40E+11 | 1.44E+14     |
| Light GBM              | 1.00E+00  | 4.51E+09  | 1.87E+13      | 9.71E-01  | 4.16E+11 | 1.39E+14     |
| Gradient Boosting      | 9.95E-01  | 1.09E+11  | 1.83E+14      | 9.67E-01  | 4.70E+11 | 5.47E+14     |
| ARIMA (SARIMAX)        | 9.63E-01  | 8.42E+11  | 2.43E+12      | 9.61E-01  | 5.60E+11 | 4.14E+12     |
| AdaBoost               | 9.71E-01  | 6.36E+11  | 2.90E+13      | 9.53E-01  | 6.67E+11 | 3.03E+14     |
| KNN                    | 4.39E-01  | 1.24E+13  | 2.97E+15      | 8.01E-01  | 2.84E+12 | 7.05E+13     |
| Elastic Net            | 5.01E-01  | 1.10E+13  | 2.48E+15      | 7.29E-01  | 3.88E+12 | 1.99E+15     |
| Ridge                  | 6.76E-01  | 7.16E+12  | 2.12E+15      | 7.25E-01  | 3.93E+12 | 1.77E+15     |
| Lasso                  | 7.13E-01  | 6.35E+12  | 2.31E+15      | 7.08E-01  | 4.17E+12 | 2.39E+15     |
| Linear Regression      | 7.13E-01  | 6.35E+12  | 2.31E+15      | 7.08E-01  | 4.17E+12 | 2.39E+15     |
| SVR(RBF)               | -2.53E-01 | 2.77E+13  | 1.60E+14      | -2.75E-01 | 1.82E+13 | 1.74E+14     |

## Final model selection: Extra Trees (Best Model)

After comparing all models, Extra Trees was selected as the best performing model.

Reasons for selecting Extra Trees

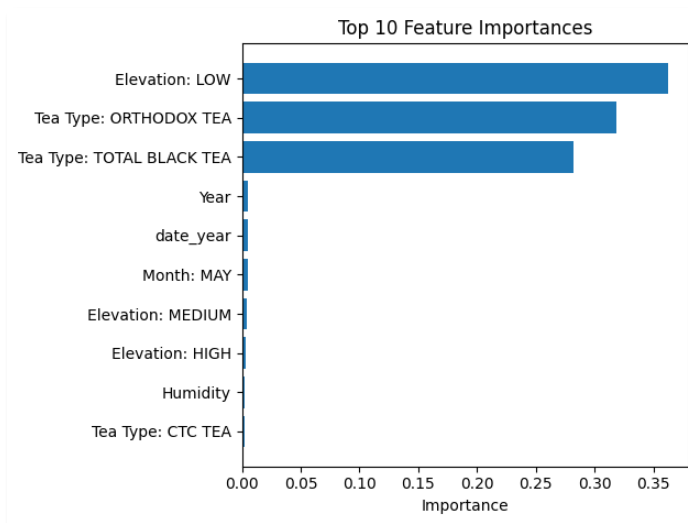
- **Best test performance overall:**
  - Test  $R^2 \approx 0.99$  (highest / among the highest)
  - Lowest test  $MSE \approx 1.48 \times 10^{11}$  compared to other candidates
- **Strong generalization**
  - Train and test scores are both very high and close to each other → indicates the model generalizes well and is not relying purely on overfitting.

- **Robustness**

- Extra Trees introduces additional randomness in tree splitting, which often improves stability and reduces variance helpful for complex, noisy patterns.

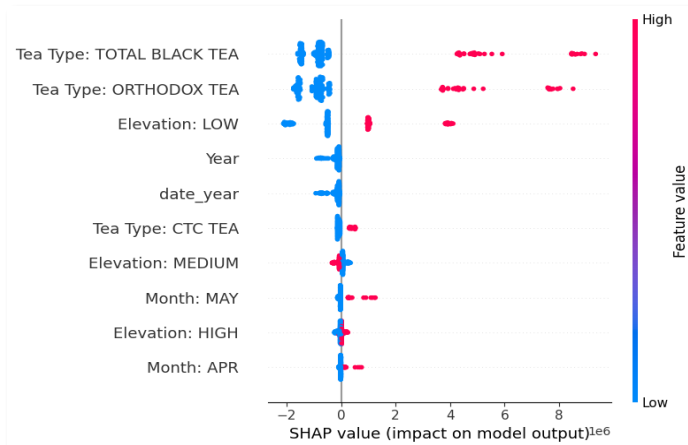
**Conclusion:** Based on highest test accuracy, lowest test error, and stable train–test behavior, **Extra Trees** was chosen as the final model for deployment/forecasting in this analysis.

*Feature importance and SHAP value graph:*



TOP 10 Feature Importances:

| Feature                   | Importance |
|---------------------------|------------|
| Elevation: LOW            | 0.362260   |
| Tea Type: ORTHODOX TEA    | 0.319026   |
| Tea Type: TOTAL BLACK TEA | 0.281759   |
| Year                      | 0.005304   |
| date_year                 | 0.005207   |
| Month: MAY                | 0.005158   |
| Elevation: MEDIUM         | 0.004114   |
| Elevation: HIGH           | 0.003156   |
| Humidity                  | 0.001992   |
| Tea Type: CTC TEA         | 0.001793   |



Both SHAP and Importance Graphs tell the same story: the model mainly cares about where the tea is grown and what type of tea it is.

- Elevation matters the most. Tea grown at low elevation consistently leads to higher predicted production.
- Tea type is the next biggest driver. **Total Black Tea** and **Orthodox Tea** are strongly associated with higher production levels.

*These two factors dominate the model's decisions.*

What matters less (but still helps a little)

- Time-related factors (such as the year) make moderate adjustments, likely capturing long-term trends.
- Seasonal and weather variables (month, rainfall, humidity, air temperature) have only a small and inconsistent influence. They fine-tune the prediction but do not change it much on their own.

## Summary

- The report analyzes and predicts Sri Lanka's tea production by integrating **temporal patterns, climatic factors (rainfall, humidity, temperature), elevation levels, and tea types**, using both exploratory analysis and predictive modeling.
- Tea production shows **strong seasonality**, peaking mainly from **March to June** and dipping in **January to February**, with an overall **declining long-term trend** observed across years.
- Among climate variables, **air temperature** has the strongest positive impact on tea production, followed by **humidity**, while **rainfall shows minimal direct influence**.
- Tea production differs significantly by elevation. **Low-elevation areas produce the highest quantities**, while **high-elevation areas show more stable but lower production**, with greater sensitivity to temperature and humidity changes.
- Production is dominated by Total Black Tea and Orthodox Tea, which also respond positively to higher temperatures. Other tea types (Green, Bio, Instant, Reclaimed) show weaker or inverse climate responses and contribute much less to total output.
- Several models were tested, and the **Extra Trees model** performed best, achieving **very high predictive accuracy (Test  $R^2 \approx 0.99$ )** with strong generalization.
- Feature importance and SHAP analysis reveal that **elevation and tea type are the most influential factors**, while climate and seasonal variables play a secondary, fine-tuning role.

## Codes

- [https://colab.research.google.com/drive/1AajgRx5V\\_GUKTk2Z9aanxZp9WsnCrzO?usp=sharing](https://colab.research.google.com/drive/1AajgRx5V_GUKTk2Z9aanxZp9WsnCrzO?usp=sharing)
- <https://drive.google.com/file/d/1-323OQaWmrvgDKExSFyNqUvDbwiAIY40/view?usp=sharing>
- [https://colab.research.google.com/drive/1V1ntuRUz7gQqeFXhBU3J\\_fkM48nKb\\_KP](https://colab.research.google.com/drive/1V1ntuRUz7gQqeFXhBU3J_fkM48nKb_KP)

## Dataset

- Train : <https://drive.google.com/file/d/1KNzLzC4omWMomEKE8MPvpz6eoTef8ifJ/view?usp=sharing>
- Validate : [https://drive.google.com/file/d/1WvYfy7jOKbmR6oy6lXI\\_FsGs0TFRjmqI/view?usp=sharing](https://drive.google.com/file/d/1WvYfy7jOKbmR6oy6lXI_FsGs0TFRjmqI/view?usp=sharing)
- Test : <https://drive.google.com/file/d/1jIRaiBMTDsZEsPQzMfSxWqMim5-dpAX3/view?usp=sharing>